SECTION 1

Q.1 A particle of mass 2 kg moves under a force F = 3x N. Find its acceleration at x = 4 m.

Q.2 Two blocks of mass 1 kg and 2 kg are connected by a string over a frictionless pulley.

Q.3 A car accelerates uniformly from rest to 20 m/s in 5 seconds. Calculate the distance covered.

SECTION 2

Q.1 State and explain Newton's second law of motion with an example.

Q.2 Define specific heat capacity and give its SI unit.

Q.3 Explain the working of a simple pendulum with a diagram.